How the Wellness Assessment App Calculates Its Results

Scoring reference for the dashboard webapp · 26 October 2026

The app turns seven measurements into a single well-being index from 0 to 100. Each measurement gets a sub-score (0–100), the sub-scores are combined with fixed weights, and a few rules subtract penalties. Everything is computed on the server (`dashboard/backend/app/scoring.py`) using the constants in `reference.py`; the page only displays the results.

Inputs and what each one affects

Input	Affects
The 7 measurement sliders	Their own sub-score, the index, the comparison table, the radar chart and the summary
Biological sex	Only the hydration target, and through it the hydration sub-score
Age, Workout minutes	Not used in the score at the moment
Dataset dropdown	Only the comparison table and the “Dataset mean” shown under each slider. The index does not use the dataset.
Examples (Non-athletic, Athlete) and Reset	Only set slider values. Reset returns every slider to its starting value, the median of the combined datasets.

1. Well-being index (“N / 100 Index”)

```
index = Σ (sub-score × weight) − penalties

The result is limited to 0–100 and displayed rounded to a whole number.
```

Sub-score	Weight
Sleep	0.20
Stress	0.15
Steps	0.15
Resting heart rate	0.15
Endurance	0.15
BMI	0.10
Hydration	0.10

Penalties

These rules are not shown on the page, but they still change the index.

Rule	Condition	Effect on the index
Underweight	BMI < 18	−4
Obesity	BMI > 30	−(3.5 + 0.35 · (BMI − 30)), at most −10
Overexertion	steps ≥ 20,000	−2
Short sleep	sleep < 6 h	−4
Long sleep	sleep > 10 h	−3
Burnout	sleep < 6.2 h **and** stress > 6	−min(15, 12 · a · b + 3.5), where a = (6.2 − sleep) / 1.6 and b = (stress − 6) / 4

2. Sub-scores (each 0–100)

The length of each bar is the sub-score. Bars are red below 50, amber from 50 to 69 and blue from 70 upwards. In the formulas, *x* is the slider value.

Metric	100 points when	Outside that range
Sleep	7–9 h	Below: $100 - 45 \cdot ((7 - x) / 1.5)^{1.4}$ Above: $100 - 50 \cdot ((x - 9) / 1.5)^{1.4}$ Minimum 0
Stress	2–3	Below 2: 95 Above 3: $100 - 92 \cdot (x - 3) / 7$, minimum 5
Steps	7,000–10,000	Below: $100 - 78 \cdot ((7000 - x) / 5000)^{1.25}$, minimum 15 Above: $100 - 16 \cdot (x - 10000) / 12000$, minimum 82
Resting heart rate	≤ 60 bpm	$100 - 8.5 \cdot ((x - 60) / 5.5)^{1.2}$, minimum 10
Endurance	≥ 18	x ≤ 4: 15 Between 4 and 18: $15 + 85 \cdot ((x - 4) / 14)^{1.05}$
BMI	22–25	Above 25: $25 + 75 / (1 + ((x - 25) / 8)^{1.7})$ 18–22: $100 - 35 \cdot (22 - x) / 4$, minimum 65 Below 18: $100 - 65 \cdot (22 - x) / 4$, minimum 15
Hydration	Inside the target band: Male 3.4–4.1 L Female 2.4–3.1 L All 2.7–3.5 L	Below the band: $100 \cdot e^{-0.45 \cdot (\text{gap} / 1.5)^{1.6}}$, minimum 10 Above the band: $100 \cdot e^{-0.35 \cdot (\text{gap} / 1.5)^{1.5}}$, minimum 35 gap = distance from the nearest edge of the band

3. Comparison with data (table)

The table compares each value with the dataset selected in the dropdown, using that dataset's mean and standard deviation (SD).

- **Value:** the slider value.
- **Dataset Mean:** the dataset's mean for that metric. The same number appears as "Dataset mean" under each slider.

- **Z-Score:** (value − mean) / SD.
- **Population %:** the normal-distribution percentile of the z-score, Φ(z) · 100. For heart rate and stress it uses Φ(−z), because lower is better; the number then means “the share of people you score better than”, so higher is always better. This assumes the metrics are normally distributed, which they are not exactly.

4. Sub-score balance (radar chart)

Each of the seven axes shows one sub-score divided by 100, drawn as a distance from the centre. Values below 10 are drawn at 10 so the shape stays visible. The chart shows the same information as the sub-score bars in a different form.

5. Summary (habit recommendation)

The app tries fourteen single changes, one at a time, and recalculates the full index for each (including penalties):

Metric	Changes tried
Sleep	+1 h, −1 h
Stress	+1, −1
Steps	+2,000, −2,000
Hydration	+0.5 L, −0.5 L
Resting heart rate	+5 bpm, −5 bpm
Endurance	+2, −2
BMI	+1, −1

Slider changes are kept within each slider’s range. The change with the largest gain is shown together with its gain in points (on a tie, the one listed first). If no change gains more than 0.4 points, the summary says “Well Balanced” instead.

Worked example: the “Non-athletic” example (index 39)

Metric	Value	Sub-score	Weight	Contribution
Sleep	5.2 h	41.9	0.20	8.38
Stress	8.5	27.7	0.15	4.16
Steps	4,200	62.2	0.15	9.33
Resting heart rate	84 bpm	50.2	0.15	7.53
Endurance	6.8	30.7	0.15	4.60
BMI	27.5	90.9	0.10	9.09
Hydration (All)	1.6 L	76.0	0.10	7.60
Weighted sum				**50.70**

1. **Short sleep penalty** (sleep < 6 h): −4.00.
2. **Burnout penalty** (sleep < 6.2 h and stress > 6): a = (6.2 − 5.2) / 1.6 = 0.625, b = (8.5 − 6) / 4 = 0.625, so −(12 · 0.625 · 0.625 + 3.5) = −8.19.
3. **Index:** 50.70 − 4.00 − 8.19 = 38.5. The page shows **39 / 100**.
4. **Summary:** of the fourteen changes, +1 h of sleep gains the most (+20.1 points), because it raises the sleep sub-score and removes the short-sleep penalty.

Where the numbers come from

- **Dataset 1 statistics** come from `health_fitness_dataset.csv`. Our exploratory analysis found this dataset to be synthetic.
- **Combined and Dataset 2 statistics** were hard-coded in the original HTML prototype and do not match the Dataset 2 file (`Fitness_Health_Tracking_Dataset_with_Missing_Values.xlsx`), which also looks synthetic: none of its columns can be predicted from the others.
- **Slider starting values** (the medians of the combined datasets) are computed from both data files by `scripts/dataset_stats.py`.
- **Thresholds and curve shapes** are hand-tuned. The code comments cite literature ranges (for example 7–9 h of sleep and 7,000–10,000 steps), but the curves were not fitted to any data.